

Ubiquitous Sensing, Computing & Communication Laboratory

Quality Laboratory Manual

Experiment No. 03

**To Understand interfacing of Ultrasonic Sensor and measure
Distance Using Raspberry Pi 4 Model B.**

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Title of Experiment: To Understand interfacing of Ultrasonic Sensor and measure Distance Using Raspberry Pi 4 Model B.

Aim of Experiment: To interface Ultrasonic Sensor using Raspberry Pi 4 Model B and Python programming.

System Requirements – Raspberry pi 4 Board Win 10 and above OS, 4GB RAM, 2.33 GHz Processor.

Hardware:

- Raspberry Pi 4 Model B
- MicroSD card (with Raspberry Pi OS installed)
- 1K,2K resistor
- HC-SR04 ultrasonic sensor
- Breadboard
- Jumper wires
- Power supply (5V, 3A USB-C adapter)

Software:

- Raspberry Pi OS
- Python 3
- RPi. GPIO library

Experiment Objectives:

Objectives:

1. **Understand the Working Principle of an Ultrasonic Sensor** – Learn how ultrasonic waves are used for distance measurement and how the sensor transmits and receives signals.
2. **Interface the Ultrasonic Sensor with Raspberry Pi 4 Model B** – Establish a proper hardware connection and configure GPIO pins for communication.
3. **Implement Distance Measurement using Python** – Write a Python script to trigger the sensor, process the received signal, and calculate the distance.
4. **Analyze and Validate Measured Distance** – Compare the measured distance with actual values to evaluate accuracy and reliability.

Experiment Outcomes:

- **Successful Interfacing of the Ultrasonic Sensor with Raspberry Pi** – Properly connect and configure the sensor with Raspberry Pi 4 Model B.
- **Real-time Distance Measurement** – Measure distances using Python programming and display the output on the terminal or screen.
- **Understanding GPIO Handling in Raspberry Pi** – Gain hands-on experience in using Raspberry Pi's GPIO pins for sensor communication.
- **Application in Embedded Systems and Robotics** – Develop a foundation for integrating ultrasonic sensors in real-world applications like obstacle detection and autonomous navigation.

1.Theory:

One of the most commonly used sensors in robotics projects is the distance sensor. The Ultrasonic sensor HC-SR04 can easily help the robot navigate around in a controlled environment. It can also be used to calculate the distance to an object in front of it. This Ultrasonic sensor is a very inexpensive electronic component that can be used for a multitude of applications. It is used in almost all robotics projects as almost all of them are either required to measure the distance or for object intrusion. It is generally used with Arduino or other microcontrollers which provide 5V.

However, this Ultrasonic sensor can also be used with Raspberry Pi. As the Ultrasonic sensor requires 5 volts, we will need to connect a couple of resistors to interface it with the 3.3V GPIO pins. The Ultrasonic sensor can measure from about 2 cm to 400 cm. And it draws very little current so it can be used with battery-powered devices.

This Ultrasonic sensor works on the principle of SONAR.

SONAR (sound navigation and ranging) is very similar to the RADAR system. In both of the systems, the detection is based on the propagation of waves between the target and detector. There are two types of sonar systems, first is active sonar systems. Here the wave propagates from the transmitter to the target and back to the receiver, which is similar to pulse-echo radar. Then there are passive sonar systems. Here the target is the source of the energy which propagates to the receiver, which is similar to passive infrared detection.

3. Connections

The Ultrasonic sensor has 4 pins:

VCC pin: For Power supply.

TRIG pin: This pin is the input pin and transmits the waves.

ECHO pin: This pin is the output pin and it detects the reflected wave.

GND pin: This pin is the ground pin.

4. Setting up the circuit

The **HC-SR04 ultrasonic sensor** is a widely used distance measurement module that works on the principle of **echo reflection**. It transmits an ultrasonic pulse (sound wave) and calculates the distance by measuring the time taken for the echo to return after reflecting from an object.

When a trigger pulse of $10\mu\text{s}$ is applied to the Trig pin, the sensor emits an ultrasonic sound wave at 40 kHz frequency. The sound wave travels through the air and reflects back if it encounters an obstacle.

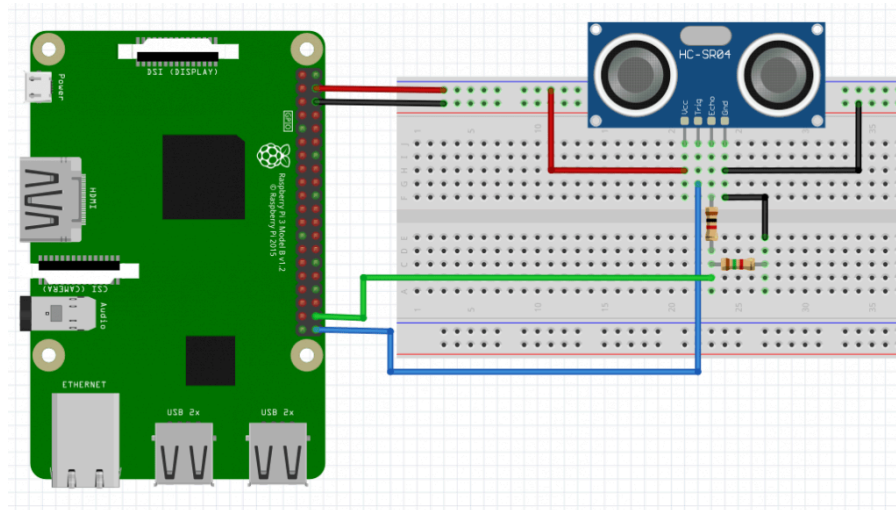
The reflected wave is detected by the Echo pin, which then outputs a high signal for a duration corresponding to the time taken for the sound wave to travel to the obstacle and return.

The distance is calculated using the formula:

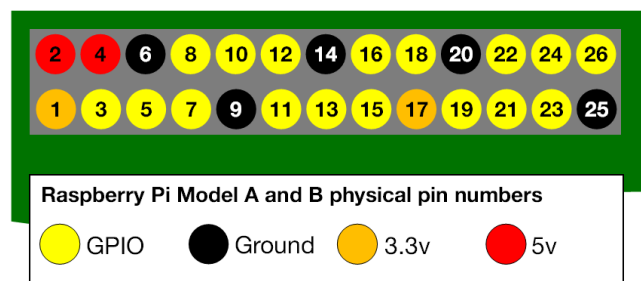
$$\text{Distance} = (\text{Time} \times \text{Speed of Sound}) / 2$$

where the **speed of sound** in air is **343 m/s (or 0.0343 cm/ μs)**.

A voltage divider circuit using $1\text{k}\Omega$ and $2\text{k}\Omega$ resistors is placed between the Echo pin and Raspberry Pi to step down the 5V signal to 3.3V.

Circuit Diagram:-

To find the pin number refer to this diagram showing the physical pin numbers on the Raspberry Pi.

**5. Python Code for Ultrasonic Sensor Interfacing with Raspberry Pi**

```
import RPi.GPIO as GPIO
import time
TRIG=21
ECHO=20
GPIO.setmode(GPIO.BCM)
while True:
    print"distance measurement in progress"
    GPIO.setup(TRIG,GPIO.OUT)
    GPIO.setup(ECHO,GPIO.IN)
    GPIO.output(TRIG,False)
    print"waiting for sensor to settle"
    time.sleep(0.2)
    GPIO.output(TRIG,True)
    time.sleep(0.00001)
```

```
GPIO.output(TRIG,False)
while GPIO.input(ECHO)==0:
    pulse_start=time.time()
while GPIO.input(ECHO)==1:
    pulse_end=time.time()
pulse_duration=pulse_end-pulse_start
distance=pulse_duration*17150
distance=round(distance,2)
print"distance:",distance,"cm"
time.sleep(2)
```

6.Observations:-

1. **Surface Type Affects Accuracy** – Hard, flat surfaces give accurate readings, while soft or irregular surfaces cause errors.
2. **Environmental Impact** – Temperature and humidity affect the speed of sound, slightly altering distance measurements.
3. **Stable Readings Need Proper Timing** – Correct trigger pulse and a small delay (e.g., 50ms) between readings improve accuracy.

7.Conclusion:-

This circuit allows safe and accurate measurement of distance using the **HC-SR04 ultrasonic sensor** and **Raspberry Pi 4 Model B**. The Raspberry Pi **triggers the sensor**, reads the **echo pulse duration**, and calculates the **distance**, which can be displayed on a screen or used in robotic applications.

Expected Oral Questions:

1. How does the HC-SR04 ultrasonic sensor measure distance?
2. Why do we use a voltage divider for the Echo pin when connecting to Raspberry Pi?
3. What is the formula used to calculate distance from the sensor readings?
4. How do environmental factors like temperature and humidity affect ultrasonic distance measurement?
5. Why is a delay needed between consecutive distance measurements?

FAQs in Interview:

1. How does an ultrasonic sensor work, and what principle does it use for distance measurement?
2. Why is it necessary to use a voltage level shifter or divider when interfacing HC-SR04 with Raspberry Pi?
3. What are the advantages and limitations of using an ultrasonic sensor for distance measurement?
4. Can you explain the role of GPIO pins in Raspberry Pi for interfacing with sensors like HC-SR04?
5. How would you improve the accuracy and reliability of distance measurements in an embedded system using an ultrasonic sensor?